KAIZEN MATHS

Lesson starter

Newton's Second Law: F = ma

Forces in One Dimension

Name

S1

A car of mass 1200 kg has driving force 1700 N and resistance 500 N. Find its acceleration.

S2

A trolley of mass 30 kg accelerates at 0.5 m s^{-2} against a resistance of 15 N. Find the driving force.

S3

A block of mass 12 kg is pulled along a rough horizontal surface. The coefficient of friction is 0.25. Find the pulling force needed to produce an acceleration of 1 m s^{-2}.

S4

A particle of mass 1 kg is lifted vertically by a string with acceleration 0.8 m s^{-2} upwards. Find the tension in the string. Take g = 9.8 m s^{-2}.

Show your working

KAIZEN MATHS

Lesson starter

Newton's Second Law: F = ma

Forces in One Dimension

Name

S1

A car of mass 1200 kg has driving force 1700 N and resistance 500 N. Find its acceleration.

S2

A trolley of mass 30 kg accelerates at 0.5 m s^{-2} against a resistance of 15 N. Find the driving force.

S3

A block of mass 12 kg is pulled along a rough horizontal surface. The coefficient of friction is 0.25. Find the pulling force needed to produce an acceleration of 1 m s^{-2}.

S4

A particle of mass 1 kg is lifted vertically by a string with acceleration 0.8 m s^{-2} upwards. Find the tension in the string. Take g = 9.8 m s^{-2}.

MEC-BF46-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Newton's Second Law: F = ma

Forces in One Dimension

Name _____

EA1

A car of mass 1200 kg has driving force 2040 N and resistance 600 N. Find its acceleration.

EA2

A trolley of mass 80 kg accelerates at 1.5 m s^{-2} against a resistance of 20 N. Find the driving force.

Show your working

KAIZEN MATHS

Exit ticket

Newton's Second Law: F = ma

Forces in One Dimension

Name _____

EB1

A block of mass 12 kg is pulled along a rough horizontal surface. The coefficient of friction is 0.2. Find the pulling force needed to produce an acceleration of 1.2 m s^{-2}.

Show your working

KAIZEN MATHS

Exit ticket

Newton's Second Law: F = ma

Forces in One Dimension

Name _____

EA1

A car of mass 1200 kg has driving force 2040 N and resistance 600 N. Find its acceleration.

EA2

A trolley of mass 80 kg accelerates at 1.5 m s^{-2} against a resistance of 20 N. Find the driving force.

MEC-BF46-L2 · EA

Show your working

KAIZEN MATHS

Exit ticket

Newton's Second Law: F = ma

Forces in One Dimension

Name _____

EB1

A block of mass 12 kg is pulled along a rough horizontal surface. The coefficient of friction is 0.2. Find the pulling force needed to produce an acceleration of 1.2 m s^{-2}.

MEC-BF46-L2 · EB

Show your working

Teacher answers & working

Newton's Second Law: F = ma · Forces in One Dimension

MEC-BF46-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

A car of mass 1200 kg has driving force 1700 N and resistance 500 N. Find its acceleration.

Answer:

$1\ m\ s^{-2}$

Working:

Take the direction of motion as positive.

Resultant force = driving force - resistance

Resultant force = 1700 - 500

Resultant force = 1200 N

$F = ma$

$1200 = 1200a$

Divide both sides by the mass.

$a = \frac{1200}{1200}$

$a = 1\ m\ s^{-2}$

Acceleration = 1 m s^{-2}.

S2

A trolley of mass 30 kg accelerates at 0.5 m s^{-2} against a resistance of 15 N. Find the driving force.

Answer:

30 N

Working:

Take the direction of acceleration as positive.

Resultant force = driving force - resistance

Resultant force = ma

Driving force - 15 = 30 × 0.5

Driving force - 15 = 15

Driving force = 15 + 15

Driving force = 30 N

Driving force = 30 N.

Teacher answers & working

Newton's Second Law: F = ma · Forces in One Dimension

MEC-BF46-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

A block of mass 12 kg is pulled along a rough horizontal surface. The coefficient of friction is 0.25. Find the pulling force needed to produce an acceleration of 1 m s^{-2}.

Answer:

41.4 N

Working:

On a horizontal surface, the normal reaction is equal to the weight.

$R = mg = 12 \times 9.8 = 117.6$ N

Friction = μR = 0.25 × 117.6 = 29.4 N

Take the direction of motion as positive.

Pulling force - friction = ma

Pulling force - 29.4 = 12 × 1

Pulling force = 12 + 29.4

Pulling force = 41.4 N

Pulling force = 41.4 N.

S4

A particle of mass 1 kg is lifted vertically by a string with acceleration 0.8 m s^{-2} upwards. Find the tension in the string. Take g = 9.8 m s^{-2}.

Answer:

10.6 N

Working:

Take upward as positive because the acceleration is upward.

Forces upward: T

Forces downward: mg

Resultant force upward = T - mg

$T - mg = ma$

$T - (1 \times 9.8) = 1 \times 0.8$

$T - 9.8 = 0.8$

$T = 9.8 + 0.8 = 10.6$ N

Tension = 10.6 N.

Teacher answers & working

Newton's Second Law: F = ma · Forces in One Dimension

MEC-BF46-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

A car of mass 1200 kg has driving force 2040 N and resistance 600 N. Find its acceleration.

Answer:

$1.2\ \text{m s}^{-2}$

Working:

Take the direction of motion as positive.

Resultant force = driving force - resistance

Resultant force = 2040 - 600

Resultant force = 1440 N

$F = ma$

$1440 = 1200a$

Divide both sides by the mass.

$a = \frac{1440}{1200}$

$a = 1.2\ \text{m s}^{-2}$

Acceleration = 1.2 m s^{-2}.

EA2

A trolley of mass 80 kg accelerates at 1.5 m s^{-2} against a resistance of 20 N. Find the driving force.

Answer:

$140\ \text{N}$

Working:

Take the direction of acceleration as positive.

Resultant force = driving force - resistance

Resultant force = ma

Driving force - 20 = 80 × 1.5

Driving force - 20 = 120

Driving force = 120 + 20

Driving force = 140 N

Driving force = 140 N.

Teacher answers & working

Newton's Second Law: F = ma · Forces in One Dimension

MEC-BF46-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

A block of mass 12 kg is pulled along a rough horizontal surface. The coefficient of friction is 0.2. Find the pulling force needed to produce an acceleration of 1.2 m s^{-2}.

Answer:

37.92 N

Working:

On a horizontal surface, the normal reaction is equal to the weight.

$R = mg = 12 \times 9.8 = 117.6$ N

Friction = μR = 0.2 × 117.6 = 23.52 N

Take the direction of motion as positive.

Pulling force - friction = ma

Pulling force - 23.52 = 12 × 1.2

Pulling force = 14.4 + 23.52

Pulling force = 37.92 N

Pulling force = 37.92 N.